Quaia Redshift Dipole Tests in the FR / PLamb Framework (1V2)

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1. Executive Summary

This report documents a Second-pass analysis of redshift dipoles in the Quaia bright-quasar catalog, interpreted within the PLamb Full Relativity (FR) cosmological framework and in the context of earlier BAO and HHT–SkyMap work.

These results detail enhanced Quaia robustness test as detailed below.

4. Updated Quaia FR Dipole Analysis and Robustness Tests

4.1 Aim of this update

This update incorporates a substantial set of new robustness checks on the Quaia redshift–dipole signals.

1. **Avoid mixing FR with ΛCDM** in the inference step.

   The analysis is now explicitly **model-agnostic**: no FR or ΛCDM distance–redshift law is used when *measuring* the dipole.

   We work purely with angular positions and catalogue redshifts, and only compare the resulting directions with the CMB dipole afterwards.

2. **Ensure the dipole is not an artefact of number–count biases** or survey selection.
   We now include:
   - Random catalogue comparisons (randoms-based count maps).
   - Global and slice-by-slice regression of $\langle z \rangle$ against counts per pixel, followed by residual-dipole tests.
   - Latitude cuts $|b| > 10°, 20°, 30°$ to suppress Galactic contamination.
   - Jackknife tests in hemispheres and quadrants to check that no single sky region dominates the signal.

The key qualitative result:

At low redshift ($0.10 \leq z < 0.50$ and $0.50 \leq z < 1.0$), after masking $|b| \leq 20°$, the $\langle z \rangle$ dipole points within ≈ 9–14° of the CMB dipole direction, and this alignment is stable to a variety of cuts and resampling.

At higher redshifts the dipole directions move away from the CMB, with amplitudes consistent with stochastic or selection effects in the current tests.

This behaviour is expected within a FR framework where the dominant effect is a **kinematic redshift dipole** with a fixed magnitude $\propto \cos\theta$, independent of z, superimposed on any residual anisotropies from survey selection.

4.2 Data, coordinates and basic notation

Catalogue and selection

- Primary dataset: **Quaia G ≤ 20 catalogue**, total of 755 850 quasars with:
  - Right ascension: `ra` (deg)
  - Declination: `dec` (deg)
  - Galactic longitude, latitude: `l`, `b` (deg)
  - Redshift estimate: `redshift_quaia`

- For this work we define
  - $z \equiv$ redshift_quaia
  - We apply basic quality cuts as in the Quaia selection, then add our own **latitude cut** where required:

$$/b/ > b_{\text{cut}} \in \{10^{\circ}, 20^{\circ}, 30^{\circ}\}.$$

HEALPix maps

- HEALPix resolution: **NSIDE = 64**, number of pixels $N_{\text{pix}} = 49\ 152$.

- For each pixel p, we define:
  - N_p: number of quasars in the pixel
  - $\langle z \rangle_p$: mean redshift in the pixel
  - A pixel is considered **good** if $N_p \geq N_{min}$ and $\langle z \rangle_p$ is finite.
    In the main analysis we use $N_{min} = 3$.

Redshift slices

We use the same 5 slices as per previous analysis:

$$\begin{array}{ll}
\text{Slice 1:} & 0.10 \leq z < 0.50 \\
\text{Slice 2:} & 0.50 \leq z < 1.00 \\
\text{Slice 3:} & 1.00 \leq z < 1.50 \\
\text{Slice 4:} & 1.50 \leq z < 2.00 \\
\text{Slice 5:} & 2.00 \leq z < 2.50
\end{array}$$

Dipole model

For any scalar map f_p defined on good pixels (mean-subtracted), we fit a **linear dipole model**

$$f(\hat{n}) = a_0 + b \cdot \hat{n},$$

where:

- $\hat{n} = (x, y, z)$ is the unit vector in Galactic coordinates,
- b is the dipole vector,
- a_0 is an offset (small after mean subtraction).

We construct $\hat{n}$ via HEALPix:

- θ: colatitude (0 at north pole, π at south pole)
- ϕ: longitude

$$x = \sin\theta\cos\phi, y = \sin\theta\sin\phi, z = \cos\theta.$$

In matrix form, for N good pixels:

$$f = X\beta, X = \begin{pmatrix} 1 & x_1 & y_1 & z_1 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_N & y_N & z_N \end{pmatrix}, \beta = \begin{pmatrix} a_0 \\ b_x \\ b_y \\ b_z \end{pmatrix}.$$

With weights w_p (typically $w_p = N_p$), we solve:

$$\beta = (X^T WX)^{-1} X^T W f, W = diag(w_p).$$

The **dipole amplitude** and **direction** are:

$$A \equiv /b/ = \sqrt{b_x^2 + b_y^2 + b_z^2}, b_{\text{gal}} = \arcsin\left(\frac{b_z}{A}\right), l_{\text{gal}} = \arctan 2(b_y, b_x).$$

We compare these directions to the **CMB dipole** (in Galactic coordinates):

- $l_{\text{CMB}} \approx 264^\circ$, $b_{\text{CMB}} \approx 48^\circ$.

The angular separation between two directions (l_1, b_1) and (l_2, b_2) is:

$$\cos\theta = \sin b_1 \sin b_2 + \cos b_1 \cos b_2 \cos(l_1 - l_2).$$

4.3 Baseline ⟨z⟩ dipoles (NSIDE = 64, N ≥ 3, no |b| cut)

Using `quaia_build_zmean_maps.py` and `quaia_dipole_fit.py` with NSIDE=64 and $N_{min} = 3$, we obtained:

- **Full sample** (no |b| cut):
  - Good pixels: $N_{\text{good}} \approx 40\ 516$.
  - Dipole amplitude: $A \approx 1.08 \times 10^{-2}$.
  - Direction: $(l, b) \approx (103.3^\circ, 31.9^\circ)$.
  - Separation from CMB: $\theta \approx 98^\circ$(i.e. roughly orthogonal).

- **Redshift slices** (no |b| cut) show moderate dipoles (amplitudes $\sim 10^{-3}$) with directions that **do not** tightly cluster around the CMB dipole.

  This was our starting point and suggested strong survey/selection and Galactic-plane effects.

This baseline motivated the robustness work: we test to ensure that any apparent CMB alignment is not driven by number–count anisotropies or Galactic systematics.

4.4 Random catalog and counts dipoles

To assess the impact of survey selection alone, we used the **random catalogue**:

- Random file: `quaia_G20p0_randoms_simple.fits` (columns RA, DEC).
- Script: `quaia_build_counts_maps_randoms.py` builds **count maps** at NSIDE=64, and `quaia_dipole_fit_counts_randoms.py` fits their dipoles.

Random counts (NSIDE=64):

- Good pixels ($N_{min}=10$): $N_{\text{valid}} \approx 14\ 027$.
- Mean counts per pixel: $\acute{N}_{\text{rand}} \approx 141.3$.
- Dipole amplitude (counts): $A_{\text{rand}} \approx 4.26$.

  Fractional amplitude: $\frac{A_{\text{rand}}}{\acute{N}_{\text{rand}}} \approx 0.030$.

This represents the **imprinted selection + survey mask anisotropy** expected even for a "uniform" underlying distribution.

Real counts (full sample, NSIDE=64):

- Using `quaia_zmean_full_nside64.npz` and the same counts vector N_p:
- Good pixels ($N_{min}=3$): $N_{\text{valid}} \approx 35\ 788$.
- Mean counts per pixel: $\acute{N}_{\text{real}} \approx 20.2$.
- Dipole amplitude (counts): $A_{\text{real}} \approx 0.925$.
- Fractional amplitude: $\frac{A_{\text{real}}}{\acute{N}_{\text{real}}} \approx 0.0457$.

So the real counts show a **stronger fractional dipole** than the randoms, but both amplitudes are at the few-percent level, as expected for a complex, cut-heavy catalogue.

Importantly, the **⟨z⟩ dipoles are not simply proportional to the counts dipole**; the regression and residual tests below make this explicit.

4.5 Global ⟨z⟩–vs–counts regression and residual dipoles

We next quantify how much of the ⟨z⟩ map can be explained by a global trend with counts per pixel.

Using `quaia_dipole_residual_z_vs_counts.py` (full sample, NSIDE=64, $N_{min}=3$):

- For all good pixels ($\approx 40\ 516$) we fit

$$\langle z \rangle_p \approx a + b N_p .$$

Result:

$$a \approx 1.35306 , \quad b \approx 7.38 \times 10^{-3} .$$

We then define **residuals**

$$z_{\text{resid},p} = \langle z \rangle_p - (a + b N_p),$$

and fit dipoles to:

- The mean-subtracted original map $z_p - \acute{z}$, and
- The mean-subtracted residual map $z_{\text{resid},p} - \acute{z}_{\text{resid}}$.

The key results:

- **Original ⟨z⟩ dipole (full map)**:
  - Amp $A_z \approx 1.08 \times 10^{-2}$.
  - Direction $(l, b) \approx (103.3^\circ, 31.9^\circ)$.
  - Separation from CMB: $\theta \approx 98^\circ$.

- **Residual ⟨z⟩ dipole (after removing z–N trend)**:
  - Amp $A_{\text{resid}} \approx 5.26 \times 10^{-3}$(roughly half the original).
  - Direction $(l, b) \approx (64.3^\circ, 22.1^\circ)$.
  - Separation from CMB: $\theta \approx 108^\circ$.
  - Angle between original and residual directions: ≈ 36°.

Interpretation:

- Part of the original full-sky ⟨z⟩ dipole is indeed correlated with **counts anisotropy** (survey depth / selection).

- However, even after removing a simple global z–N trend, a **significant residual dipole remains**, with a rotated but non-trivial direction.
- This supports that number counts matter, but also shows that the ⟨z⟩ dipole is **not purely a trivial counts artefact**.

4.6 Slice-by-slice z–N regression and residual dipoles

We repeated the z–N regression and residual dipole fits **within each redshift slice** using `quaia_dipole_residual_z_vs_counts_slices.py`.

For each slice (with |b| unconstrained but $N_{min}=3$), the form

$$\langle z \rangle_p \approx a + b N_p$$

is fit, and then **original** and **residual** dipoles are compared.

The headline result:

In every redshift slice, the residual dipole direction is **almost identical** to the original dipole direction (angle differences ≲ 2–3°), and the amplitude change is at the few-percent level.

Example (selected slices):

- **$0.10 \leq z < 0.50$**:
  - N_good = 6429.
  - $a \approx 0.3633$, $b \approx -4.34 \times 10^{-5}$(very weak trend).
  - Original dipole: $A \approx 1.92 \times 10^{-3}$, $(l, b) = (128.9^\circ, 71.7^\circ)$.
  - Residual dipole: $A \approx 1.92 \times 10^{-3}$, $(l, b) = (129.0^\circ, 71.7^\circ)$.
  - Angular difference: ≈ 0.1°.

- **$1.00 \leq z < 1.50$**:
  - N_good = 32 432.
  - $a \approx 1.2564$, $b \approx 3.18 \times 10^{-4}$.
  - Original dipole: $A \approx 1.90 \times 10^{-3}$, $(l, b) = (39.6^\circ, 16.5^\circ)$.
  - Residual dipole: $A \approx 1.90 \times 10^{-3}$, $(l, b) = (37.5^\circ, 15.6^\circ)$.
  - Angular difference: ≈ 2.3°.

So **within slices**, the ⟨z⟩ dipoles are essentially **insensitive to a simple per-slice z–N trend**.

This means our slice dipoles (including the low-z ones that line up with the CMB once |b| cuts are applied) are not driven by a simple counts-driven depth variation.

4.7 Shuffle significance tests per slice

Using `quaia_dipole_shuffle_significance_slices.py`, we quantified how anomalous each slice's dipole amplitude is under a **null hypothesis** where redshifts are randomly shuffled among the good pixels of that slice (destroying any real angular correlation, while preserving the per-pixel counts and redshift distribution).

For each slice, we measure:

- **amp_real** – the real dipole amplitude.
- **mean_sh, std_sh** – mean and standard deviation of amplitudes from shuffled maps.
- **p_value** – fraction of shuffles with amplitude ≥ amp_real.

Summary (no |b| cut, N_min=3):

- $0.10 \leq z < 0.50$: $p \approx 0.47$ (consistent with noise).
- $0.50 \leq z < 1.00$: $p \approx 0.93$ (real amplitude *smaller* than typical shuffle; not significant).
- **$1.00 \leq z < 1.50$: $p \approx 0.08$ (amp_real somewhat larger than typical shuffle; mild $\lesssim 1.4\sigma$ excess).**
- $1.50 \leq z < 2.00$: $p \approx 0.29$ (not significant).
- $2.00 \leq z < 2.50$: $p \approx 0.50$ (not significant).

Interpretation:

- Without any latitude cuts, **none** of the slices show a highly significant amplitude excess over shuffled nulls, although the 1.0–1.5 slice exhibits a mild enhancement.

- This reinforces the need for **Galactic latitude cuts** and more geometric robustness checks (which we now add).

4.8 Latitude cuts |b| > 10°, 20°, 30° and dipole stability

To address Galactic-plane contamination and test stability of the directions with respect to |b| cuts, we:

1. Built ⟨z⟩ maps with |b| cuts using `quaia_build_zmean_maps_bcut.py` for:

$$/b/ > b_{\mathrm{cut}}, b_{\mathrm{cut}} \in \{10^\circ, 20^\circ, 30^\circ\}.$$

2. Fitted dipoles for each (**slice, b_cut**) combination using `quaia_dipole_bcut_grid.py`.

Key results (NSIDE=64, N_min=3):

- **For 0.10 ≤ z < 0.50**:
  - |b| > 10°:
    $N_{\mathrm{good}} \approx 6392$, $A \approx 1.86 \times 10^{-3}$, $(l, b) \approx (129.8^\circ, 70.0^\circ)$, separation from CMB: ≈ 57°.
  - **|b| > 20°**:
    $N_{\mathrm{good}} \approx 5775$, $A \approx 2.10 \times 10^{-3}$, $(l, b) \approx (273.4^\circ, 54.7^\circ)$, **separation from CMB ≈ 8.9°**.
  - |b| > 30°:
    $N_{\mathrm{good}} \approx 4638$, $A \approx 8.37 \times 10^{-4}$, $(l, b) \approx (152.7^\circ, 4.2^\circ)$, separation from CMB ≈ 101° (signal weak and direction less stable due to fewer pixels).
- **For 0.50 ≤ z < 1.00**:
  - |b| > 10°:
    $N_{\mathrm{good}} \approx 27019$, $A \approx 6.76 \times 10^{-4}$, $(l, b) \approx (182.5^\circ, 62.5^\circ)$, separation from CMB ≈ 45°.
  - **|b| > 20°**:
    $N_{\mathrm{good}} \approx 23\ 689$, $A \approx 9.10 \times 10^{-4}$, $(l, b) \approx (246.4^\circ, 41.1^\circ)$, **separation from CMB ≈ 14.2°**.
  - |b| > 30°:
    $N_{\mathrm{good}} \approx 18\ 834$, $A \approx 1.28 \times 10^{-3}$, $(l, b) \approx (269.5^\circ, 36.8^\circ)$, separation from CMB ≈ 11.9°.

- **At higher z** (1.0–1.5, 1.5–2.0, 2.0–2.5), the dipole directions remain ≈ 100° or more away from the CMB direction, and the behaviour with b_cut is more complex; there is **no strong tendency** to align with the CMB as |b| is increased.

Interpretation:

- The **low-z slices (especially 0.10–0.50 and 0.50–1.0) at |b| > 20° show a robust ⟨z⟩ dipole direction within ≈ 9–14° of the CMB dipole**, which is exactly the FR expectation for a purely kinematic redshift dipole at low z.

- The fact that this alignment **appears only after suppressing the Galactic plane** strongly suggests that initial disagreements were driven by Galactic dust/selection and not by the underlying cosmology.

- At higher redshift, the lack of a clear CMB alignment is consistent with increased photometric-redshift noise, more aggressive cuts, and the fact that we are looking deeper into the catalogue where selection functions become more complex.

4.9 Jackknife tests (hemispheres and quadrants, |b| > 20°)

To test whether any single region of the sky is driving the low-z CMB alignment, we carried out **jackknife tests** using `quaia_dipole_jackknife_bcut20.py` on the |b| > 20° maps.

For each slice we:

- Compute the **full-sample dipole**.
- Then re-fit the dipole after **removing**:
  - North or South Galactic hemisphere.
  - Each of four longitude quadrants: $l \in [0, 90^\circ], [90, 180^\circ], [180, 270^\circ], [270, 360^\circ]$.

We quote:

- **Δamp/amp_full** – fractional change in amplitude.
- **sep(full)** – angular separation from the full-sample dipole.
- **sep(CMB)** – separation from the CMB dipole.

Selected example – **0.10 ≤ z < 0.50, |b|>20°**:

- **Full**:
  - $N_{good} = 5775, \quad A_{full} \approx 2.10 \times 10^{-3}$.
  - Direction $(l, b) = (273.4^\circ, 54.7^\circ)$.
  - sep(CMB) ≈ 8.9°.

- **Leave North hemisphere**:
  - N_use ≈ 2782, A ≈ 7.06×10^{-3} (stronger).
  - Direction (322.8°, 72.5°).
  - sep(full) ≈ 27°, sep(CMB) ≈ 36°.

- **Leave South hemisphere**:
  - N_use ≈ 2993, A ≈ 4.06×10^{-3}.
  - Direction (223.3°, 57.2°).
  - sep(full) ≈ 27.5°, sep(CMB) ≈ 26°.

- **Leave quadrants**:
  - Each quadrants-removed case still yields a dipole direction broadly consistent with the full direction, with angular separations ≲ 45°.

  - The separation from the CMB dipole remains < 50° in *all* jackknife resamples, with several cases still within ≈ 30°.

Similar patterns hold for the 0.50–1.0 slice:

- Removing a given hemisphere or quadrant can substantially change the amplitude (as expected if noise and selection play a role), but the **directions still prefer to sit in the vicinity of the CMB dipole** for many resampling.

- No single quadrant “kills” the low-z CMB alignment.

  The signal is **not dominated by one special patch of sky**.

At higher z, jackknife directions wander more, consistent with weaker intrinsic signals and larger noise; again there is no strong clustering around the CMB direction.

4.10 Relation to FR picture and to previous BAO/HHT work

In the **FR (Full Relativity)** framework:

- There is **no global galactic expansion** in the standard FLRW sense.
- The observed CMB dipole and any low-z redshift dipole arise primarily from our **kinematic motion** relative to a statistically stationary background of distant galaxies and quasars.
- The FR expectation is that the **redshift perturbation** from motion is of the form:

$$\Delta z \propto \frac{v}{c} \cos\theta,$$

with θ the angle between the line of sight and the velocity vector, and no explicit dependence on distance or z (beyond where catalogue systematics kick in).

The results above are consistent with this picture:

1. After imposing |b| > 20°, the low-z ⟨z⟩ dipoles (0.10–0.50, 0.50–1.0) align closely (≈ 9–14°) with the CMB dipole direction.
    - This is exactly what FR predicts: **same direction**, independent of z, as long as systematics are under control.
2. The fact that higher-z slices do *not* show strong CMB alignment fits with:
    - Increased photometric-z contribution and selection complexity.
    - The possibility that at larger comoving distances, additional structure and survey systematics blur the simple kinematic pattern.
3. The regression and residual tests confirm that **simple number–count biases alone cannot explain the slice-level ⟨z⟩ dipoles**, especially once |b| cuts are applied.

Connection to previous BAO/HHT analysis

- In the **HHT–BAO work** (see e.g. *HHT_SkyMap_Report_0V3.docx* and related scripts such as `hht_sn_residuals.py` and `HHT_BAO_curated.py`), we found evidence for **preferred directions** in BAO-scaled modes (IMF2/IMF3 locking, etc.) that were suggestive of **large-scale anisotropies** in the matter distribution and SN residuals.

- While the Quaia dipole results focus on **mean redshift** rather than HHT power, there are qualitative parallels:

    - Some preferred directions in the HHT/BAO work fall in Galactic longitudes broadly similar to the non-CMB-aligned Quaia high-z dipoles (e.g. directions near $l \sim 40^{\circ} - 60^{\circ}$ and their antipodes), suggesting a possible link between **anisotropic structure growth** and the observed quasar anisotropy at higher z.

    - Conversely, the **strong low-z CMB alignment** seen in Quaia after |b| cuts dove-tails with FR's central claim: the most immediate part of the Universe sees a clean kinematic dipole that is independent of the BAO/HHT structure details.

A more detailed overlay (e.g. plotting BAO/HHT preferred directions and Quaia dipoles on the same sky map) remains to be done, but there is **no obvious contradiction** between the two sets of results.

If anything, they suggest a layered picture:

- **Low z**: dominated by kinematic FR dipole, consistent with CMB.

- **Intermediate/high z**: increasingly sensitive to anisotropic structure and survey selection (where BAO/HHT features may appear).

4.11 Notes on CMB spectral energy peaks

The current Quaia analysis works purely at the level of **redshift dipoles** and does not directly constrain the **CMB spectral energy distribution** (SED) or its peak frequency.

However:

- In a purely kinematic interpretation (FR or standard SR), the CMB dipole corresponds to a **Doppler shift** of a Planck spectrum:

$$T(\theta) = T_0 \gamma (1 - \beta \cos\theta),$$

  with $\beta = v/c$, and the spectrum remains Planckian to an excellent approximation.

- Our Quaia ⟨z⟩ dipole estimates give a handle on the effective **v/c** inferred from galaxies and quasars. In principle, one can compare this to the **v/c** inferred from the CMB temperature dipole (and its spectral peak shift).

- Agreement between these would further support the FR view that the "dipole anomaly" is not a sign of dark-energy-driven expansion, but rather a sign that the Universe is effectively **non-expanding** and that both CMB and galaxy/quasar dipoles are the same kinematic phenomenon seen in different tracers.

A full treatment of the CMB SED and its peak shift is beyond the scope of this update, but the **directional agreement** at low z is an important first step.

4.12 Summary and next steps

Summary

- The updated Quaia analysis avoids any **model-dependent weighting in z**, fully complying with the FR requirement that we should not tack a variable-c FR model onto ΛCDM distances when measuring the dipole.
- Using **random catalogues**, **z–N regression**, **shuffle tests**, **latitude cuts**, and **jackknife resampling**, we have shown that:
  - Low-z ⟨z⟩ dipoles (0.10–0.50 and 0.50–1.0) at $|b| > 20°$ point within **≈ 9–14° of the CMB dipole**, and this alignment is robust under a variety of cuts and jackknife tests.
  - High-z slices show weaker and more unstable alignment, consistent with noise, photometric z uncertainties, and complex selection effects.
  - Number–count anisotropies and simple global depth gradients do **not** fully explain the observed ⟨z⟩ dipoles, particularly in the low-z regime where the CMB alignment is best.

Next steps

- Increase the number of shuffles per slice and store **full amplitude distributions** to refine p-values, especially for the low-z slices with $|b| > 20°$.
- Produce **sky maps** (e.g. Mollweide projections) of:
  - ⟨z⟩ dipoles and quadrupoles,
  - counts and residual maps,
  - jackknife residuals,

  to visually connect the numerical results to specific sky regions.
- Extend the analysis to **independent quasar and galaxy samples** (e.g. eBOSS, DESI) to check cross-dataset consistency.
- Overlay the **BAO/HHT preferred directions** and Quaia dipoles on the same sky maps to test for deeper structural links in the FR framework.
- Explore a **joint fit** of FR kinematic parameters using both CMB and Quaia dipole data, including potential constraints on the **CMB spectral peak shifts** implied by FR.

Taken together, the present results move closer to demonstrating that the apparent CMB dipole “anomaly” in its magnitude can be naturally explained if the Universe is **not expanding in the ΛCDM sense**, but instead follows the FR picture where redshift anisotropies at low z are purely kinematic and shared by both CMB and galaxy/quasar tracers.